Phase 1: Project Proposal & Environment Setup

Data Engineering Course | Project: Bank Customer Churn Pipeline

1. Team Members & Roles

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Note: Every team member understands the full end-to-end architecture and pipeline workflow.

2. Dataset Specification

Profile Overview

- **Dataset Name:** Bank Customer Churn (European bank customers; widely published on Kaggle as "Churn Modelling")
- **Download Source:** https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling?utm_source=chatgpt.com
- **File Format:** CSV (`Bank_Customer_Churn.csv`)
- **Volume:** 10,000 rows | 14 columns
- **Data Integrity:** 0 missing values | 0 duplicate rows

- **Target Distribution:** ~20.4% customer churn rate (`Exited = 1`), making it highly suitable for binary classification and churn risk segmentation. Meets minimum volume requirements.

Data Dictionary

Column Name	Data Type	Category	Notes & Business Rules
`RowNumber`	Integer	ID	Row index (dropped in clean analytics schema)
`CustomerId`	Integer	ID	Unique primary key for each customer
`Surname`	Text	Text	2,932 distinct customer surnames
`CreditScore`	Integer	Numeric	Score ranges from 350 to 850
`Geography`	Text	Categorical	Country of operation (France, Germany, Spain)
`Gender`	Text	Categorical	Customer gender (Male, Female)
`Age`	Integer	Numeric	Customer age (18 to 92 years)
`Tenure`	Integer	Numeric	Years as account holder (0 to 10 years)
`Balance`	Decimal	Numeric	Account balance (contains zero-balance accounts)
`NumOfProducts`	Integer	Numeric/Cat.	Number of bank products held (1 to 4)
`HasCrCard`	Integer (0/1)	Binary	Credit card ownership flag
`IsActiveMember`	Integer (0/1)	Binary	Active status flag

Column Name	Data Type	Category	Notes & Business Rules
`EstimatedSalary`	Decimal	Numeric	Estimated annual salary
`Exited`	Integer (0/1)	Target	Churn flag (`1` = Left the bank, `0` = Retained)

Timestamp Note: The source dataset lacks a temporal column. An ingestion timestamp (`ingested_at`) will be injected during the Python ingestion step for pipeline monitoring and batch execution tracking.

3. Analytical Objectives

- **Question 1**
- **Which age groups have the highest customer churn rate?**
- **Analysis: Group customers by age ranges and compare their churn rates.**
- **Question 2**
- **How does customer balance affect the probability of leaving the bank?**
- **Analysis: Compare account balances between customers who stayed and customers who left.**
- **Question 3**
- **Which geographical region has the highest customer churn rate?**
- **Analysis: Compare churn rates across France, Germany, and Spain.**
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4. End-to-End System Architecture

Pipeline Components

Pipeline Layer	Component	Description & Role
Data Source	Kaggle CSV File	Raw source file `Bank_Customer_Churn.csv`
Ingestion Engine	Python Script	Extracts CSV, validates data quality, appends `ingested_at`, loads into `raw` schema
Storage Layer	PostgreSQL DB	Stores raw ingested tables and curated analytics schemas/views
Orchestration	Kestra	Orchestrates scheduled workflow runs, data validations, and SQL models
Visualization	BI Dashboard	Connects to analytics views to visualize churn metrics and segmentations

Execution Flow & Infrastructure

1. Containerization: PostgreSQL 16 and pgAdmin 4 execute within Docker containers linked via a shared virtual network (`de_net`). Database state is persisted using a Docker volume (`pgdata`).
2. Data Flow:
    - Python scripts (`pandas` + `SQLAlchemy` / `psycopg2`) read and validate raw CSV records.
    - Cleaned records are written to the `raw.bank_customer_churn` staging table with system timestamps.

    - DDL and SQL transformation pipelines construct the `analytics` schema and aggregate analytical views.
3. Automation: Kestra orchestrates scheduled execution of python scripts, SQL transformations, and analytical data quality assertions.

5. Expected Database Schemas & Views

Schema . Table / View	Type	Purpose	Primary Key / Key Columns
`raw.bank_customer_churn`	Table	Exact copy of source CSV plus ingestion tracking metadata	`row_number`, `customer_id`, `ingested_at`
`analytics.customers`	Table	Cleaned, typed, deduplicated dimension table with strict constraints	`customer_id` (PK), `geography`, `gender`, `is_active_member` (bool), `exited` (bool)
`analytics.v_churn_by_geography`	View	Aggregate summary table addressing Question 1	`geography`, `customers`, `churn_rate_pct`, `avg_balance`
`analytics.v_churn_by_age_group`	View	Age cohort breakdown addressing Question 2	`age_group`, `customers`, `churn_rate_pct`
`analytics.v_churn_by_products_activity`	View	Cross-tabulation table addressing Question 3	`num_of_products`, `is_active_member`, `customers`, `churn_rate_pct`

6. Development Environment & Repository Layout

Infrastructure Configuration

- **Container Services:** PostgreSQL 16 (configured with automated healthchecks and volume persistence) alongside pgAdmin 4.

- **Environment Variables:** Credentials and database parameters are managed securely via a local `.env` file (excluded from source control), with a template provided in `.env.example`.

Repository Structure

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None
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├── docs/                       # Project documentation and architectural
diagrams
├── scripts/                    # Python data ingestion & cleaning scripts
├── sql/                        # DDL, DML schema definitions, and analytical
views
├── flows/                      # Kestra workflow definition files (YAML)
├── docker-compose.yml          # Container services orchestration configuration
├── .env.example                # Environment variable configuration template
└── README.md                   # Complete project setup and execution guide
```

Links & Deliverables

- **GitHub Repository:** https://github.com/mohamedmohab-20/Data-engineering-project-task1
- **Video Walkthrough:** https://drive.google.com/file/d/1HPI0RIodctf9IqHB7IkwotYnjtWfayja/view?usp=sharing